

Second submission: Feminine Narratives of Opium Use in the Long Nineteenth Century

Bich Ngoc Doan, Clémentine Martinez-Perez, Eglantine Vialaneix, Samuel Zimmermann

Literature Review

The cultural and literary history of drugs in the nineteenth and early twentieth centuries has attracted considerable scholarly attention, yet women's roles within these histories remain underexplored (Buxton 2021). Historical scholarship has traced opium's social, economic, and medical significance in the England and in the United States, documenting its widespread use, the gradual emergence of regulatory frameworks (Berridge 1979; Foxcroft 2007), and the gendered discourses surrounding drug consumption reflecting moral anxieties (Madancy 2013). Women's agency in these histories is rarely foregrounded, partly because they left fewer traces in public archives, and partly because drug consumption, like so many social practices, was shaped by a sharp division between a domestic sphere assigned to women and a public sphere dominated by men (Perrot 1998).

Where women appear, they are consistently refracted through intersecting hierarchies of class and respectability. Research on female consumption, particularly of drugs and alcohol, reveals a recurring paradox, one illustrated by Enefalk's (2015) study of Swedish women's diaries, whose logic structured female drug consumption across British and American contexts alike. Bourgeois morality increasingly condemned female consumption, yet elite women's use persisted, privatised and medicalised into respectability, while working-class intoxication, whether alcohol or opium, was coded as sexual disorder and moral failure. Visibility, not consumption itself, was what threatened the social order. As Miller and Carbone-Lopez (2012) argue, gender should be understood not as an individual attribute but as a configuration of practices within a relational system, always intersecting with class, race, and place, which constitute an essential framework for any analysis of who was permitted to consume what, how, and where.

Furthermore, the dominant narrative through which drug use was interpreted shifted considerably between the late nineteenth and early twentieth centuries. This timeframe saw the gradual consolidation of "addiction" as a medical and moral category (Delcourt 2016), one that reframed habitual consumption from a private domestic habit into a pathological, socially threatening condition (Berridge 1988; Seddon 2008). This pivot had profound gendered consequences, producing new and unstable figurations of the female drug user. The woman who self-medicated quietly at home, or who soothed and administered opiates to others as part of her domestic role, increasingly gave way to more transgressive figures, like the recreational user in urban entertainment spaces (as epitomized by scandals of opium-related deaths of actresses at the beginning of the twentieth century), whose presence could no longer be contained within the home (Carey 2014). Literary scholarship confirms this trajectory, but also reveals that the perspective of the narrator shapes how female drug use is portrayed: where male authors constructed the opium-inspired visionary or the morally suspect addict through an externalising lens, what Evans and Williams (2013) term "*the male gaze*", women writers such as Harriet Martineau and Elizabeth Barrett Browning framed drug use in therapeutic and domestic terms, foregrounding interiority and medical necessity rather than spectacle (Freeman 2012; Zieger 2008).

Yet these findings remain dispersed across isolated case studies, separated by national tradition and reliant on close reading of individual texts. Existing scholarship has developed along strictly national lines: historians and literary scholars working on British drug cultures rarely engage with their American counterparts, and vice versa, leaving the two traditions in isolation despite the shared cultural, social and pharmaceutical transformations of the period. Whether the shifting figurations of femininity attached to opium, from domestic self-medicator and caring healer to recreational user and

transgressive agent, constitute a broad cultural shift or merely isolated literary gestures remains unanswerable through close reading alone, and has never been examined across both traditions simultaneously.

Digital humanities methods offer a way to address this gap: Computational analysis of large corpora can reveal structural patterns and gradual semantic shifts invisible to individual readers. Drawing on word embedding approaches that quantify evolving gender stereotypes across literary corpora (Garg et al. 2018), and informed by gender studies frameworks, particularly the notions of agency and the male gaze, computational analysis of large corpora can reveal structural patterns and gradual semantic shifts that close reading alone cannot capture.

Research Question

How do literary constructions of femininity associated with opiates evolve between the 1870s and 1920s across British and American literature?

Sub-question: To what extent can corpus-scale analysis detect and trace shifts in the stereotypes attached to female drug users, and do any such patterns align with the broader cultural pivot in addiction narratives at the turn of the twentieth century?

Methods

Dataset Enrichment and Filtering

To study literary constructions of femininity across British and American literature at scale, we draw on the Project Gutenberg dataset, which comprises more than 75,000 digitized books alongside rich metadata. To enhance coverage and reliability, we integrate external sources, including the Chronoberg dataset (Hegde et al. 2025) and the Wikipedia API, enabling us to retrieve key attributes such as publication year and author gender.

We define a temporal scope spanning 1870 to 1920, a period that captures the high-imperial era of the Opium Wars and concludes in the consolidation of international drug prohibition in the early twentieth century. Within this interval, we filter the corpus to include British and American literary works containing references to opium-related terms, identified through a curated historical lexicon. This procedure yields a final corpus of 3,743 books.

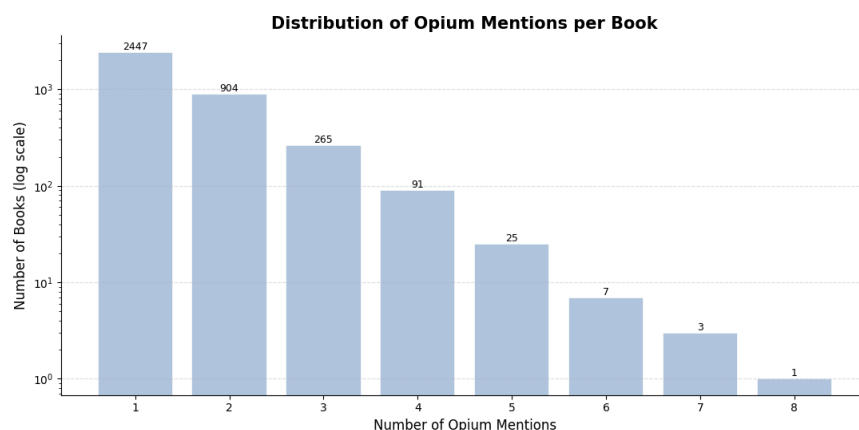


Figure 1: Counts of the number of opium-related mentions per book in American and English literary texts published between 1870 and 1920.

Analysis

Our methodology is designed to move from broad thematic patterns to the granular quantification of character agency, using a 100-word **"opium window"** as the primary unit of analysis. Preliminary validation confirmed that this window size is sufficient to capture the immediate narrative context without introducing extraneous noise from unrelated plot points.

Gender Extraction

We employ a hybrid attribution pipeline combining the precision of syntactic parsing with broader contextual recall. Using spaCy's dependency parser, we identify the grammatical subject (**nsubj/nsubjpass**) governing verbs linked to opium mentions. When gender cues fall outside the sentence, a fallback heuristic assigns gender based on the proximity and density of gendered markers within a reduced context window. First-person narration is resolved via co-reference: we distinguish between character speech and authorial voice, inferring gender from metadata where necessary.

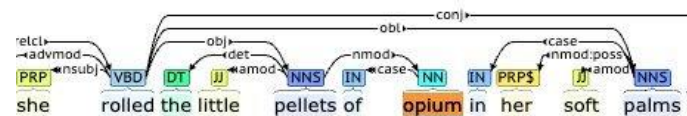


Figure 2: Example of dependency parsing: *opium* (keyword) → *rolled* (main verb) → *she* (nominal subject, representing “women”).

The Macro-Linguistic Landscape

We map the broader opium discourse through two complementary methods. First, through comparative lexicography, we isolate gendered verbs, adjectives, and nouns present in each opium window, using statistical tests (e.g., log-likelihood ratios) and visualizations (e.g. word cloud) to reveal whether distinct lexical patterns characterize female versus male-centered narratives—for instance, whether women are disproportionately associated with medical or domestic registers. This is complemented by unsupervised topic modeling (LDA (Blei et al. 2003) or BERTopic (Grootendorst 2022)), through which we identify recurring motifs across the corpus and track how their distribution shifts over time and across national contexts (British vs. American literature).

Quantifying Agency via Connotation Frames

To operationalize agency, we apply the Connotation Frame framework (Rashkin et al. 2016), previously applied to online discourse (Rashkin et al. 2017) and gendered perspectives in movies (Sap et al. 2017), which scores verbs along dimensions of agency, power, and sentiment. This distinguishes active behaviors (e.g., “sought”) from passive ones (e.g., “given”), and evaluates whether characters hold authority or are medically controlled. Temporal trends in these scores could provide empirical evidence for shifts toward the medicalization of female drug use.

The Authorial Lens: Modeling the “Gaze”

Finally, we segment results by author gender to examine narrative perspective. This enables testing of the “male gaze” and “female gaze” hypothesis, contrasting externalized, moralized portrayals with internally motivated, medically framed depictions of female opium use. Building on the methods previously presented, we evaluate systematic presence of sexualized female characters and their agency to quantify both theories.

Limitations and Challenges

A primary limitation is incomplete publication metadata in the Project Gutenberg corpus. Although we partially recover dates via external sources, coverage remains uneven, introducing potential bias.

In addition, because the dataset is predominantly American in origin, the corpus construction may skew comparisons between British and American literature.

Our analysis also depends critically on accurate gender attribution. Author gender inference may underrepresent lesser-known female writers, thereby amplifying already well-documented figures. Character-level gender extraction, largely driven by dependency parsing, is challenging to perform. When parsing fails, we fallback to heuristic methods which risk overlooking less explicit or nonstandard gender cues, potentially excluding a broader diversity of characters from subsequent analysis.

References

- Berridge, Virginia. 1988. "The Origins of the English Drug 'Scene' 1890–1930." *Medical History* 32 (1): 51-64.
- Berridge, Virginia. 1999. "Histories of Harm Reduction: Illicit Drugs, Tobacco, and Nicotine." *Substance Use & Misuse* 34 (1): 35-47.
- Blei, D. M., Ng, A. Y., and Jordan, M. I. 2003. "Latent Dirichlet Allocation." *Journal of Machine Learning Research* 3: 993–1022.
- Buxton, Julia. 2020. "Women as Actors in the Drug Economy." In *The Impact of Global Drug Policy on Women: Shifting the Needle*, edited by Julia Buxton, Giavana Margo, and Lauren Burger, 147-58. Bingley: Emerald Publishing Limited.
- Carey, Elaine. 2014. *Women Drug Traffickers: Mules, Bosses, and Organized Crime*. Albuquerque: University of New Mexico Press.
- Clark, Peter. 1983. *The English Alehouse: A Social History 1200–1830*. London: Longman.
- Delcourt, Irène. 2016. "La Noire Idole: Opium, ivresse et ébriété au XIXe siècle aux États-Unis." *Nuevo Mundo Mundos Nuevos*. <https://doi.org/10.4000/nuevomundo.69852>.
- Drug Enforcement Administration. 2018. *Slang Terms and Code Words: A Reference for Law Enforcement Personnel (DEA Intelligence Report DIR-022-18)*. U.S. Department of Justice. <https://www.dea.gov/sites/default/files/2018-07/DIR-022-18.pdf>
- Enefalk, Hanna. 2015. "Alcohol and Femininity in Sweden c. 1830–1922: An Investigation of the Emergence of Separate Drinking Standards for Men and Women." *Substance Use & Misuse* 50 (6): 736–46.
- Evans, Mary, and Williams, Carolyn. 2013. *Gender: The Key Concepts*. London: Routledge.
- "Female Gaze." Wikipedia, 6 Jan. 2026. Wikipedia, https://en.wikipedia.org/w/index.php?title=Female_gaze&oldid=1331505015.
- Fleetwood, Jennifer, and Leban, Lindsay. 2023. "Women's Involvement in the Drug Trade: Revisiting the Emancipation Thesis in Global Perspective." *Deviant Behavior* 44 (2): 238–58.
- Foxcroft, Louise. 2007. *The Making of Addiction: The 'Use and Abuse' of Opium in Nineteenth-Century Britain*. The History of Medicine in Context. Aldershot: Ashgate.
- Freeman, Hannah Cowles. 2012. "Opium Use and Romantic Women's Poetry." *South Central Review* 29 (1): 1–20.
- Garg, Nikhil, Schiebinger, Londa, Jurafsky, Dan, and Zou, James. 2018. "Word Embeddings Quantify 100 Years of Gender and Ethnic Stereotypes." *Proceedings of the National Academy of Sciences* 115 (16): E3635–E3644.
- Grootendorst, M. 2022. "BERTopic: Neural Topic Modeling with a Class-Based TF-IDF Procedure." *arXiv preprint arXiv:2203.05794*.

Gutmann, Matthew C. 1996. *The Meanings of Macho: Being a Man in Mexico City*. Berkeley: University of California Press.

Hunt, Geoffrey, Asmussen Frank, Vibeke, and Moloney, Molly. 2015. "Rethinking Gender within Alcohol and Drug Research." *Substance Use & Misuse* 50 (6): 685–92.

Madancy, Joyce A. 2013. "Smoke and Mirrors: Gender, Colonialism, and the Royal Commission on Opium, 1893–95." *The Social History of Alcohol and Drugs* 27 (1): 37–61.

Maher, Lisa, and Hudson, Susan L. 2007. "Women in the Drug Economy: A Metasynthesis of the Qualitative Literature." *Journal of Drug Issues* 37 (4): 805–26.

"Male Gaze." Wikipedia, 19 Apr. 2026. Wikipedia, https://en.wikipedia.org/w/index.php?title=Male_gaze&oldid=1349868637.

Miller, Jody, and Carbone-Lopez, Kristin. 2015. "Beyond 'Doing Gender': Incorporating Race, Class, Place, and Life Transitions into Feminist Drug Research." *Substance Use & Misuse* 50 (6): 693–707.

Perrot, Michelle. 1998. *Les Femmes ou les silences de l'Histoire*. Paris: Flammarion.

Hegde, N., Paul, S., Joel-Frey, L., Brack, M., Kersting, K., Mundt, M., and Schramowski, P. 2025. "CHRONOBERG: Capturing Language Evolution and Temporal Awareness in Foundation Models." *arXiv preprint arXiv:2509.22360*.

Rashkin, Hannah, Singh, Sameer, and Choi, Yejin. 2016. "Connotation Frames: A Data-Driven Investigation." In *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 311–321. Berlin, Germany: Association for Computational Linguistics. DOI: 10.18653/v1/P16-1030.

Rashkin, Hannah, Bell, Eric, Choi, Yejin, and Volkova, Svitlana. 2017. "Multilingual Connotation Frames: A Case Study on Social Media for Targeted Sentiment Analysis and Forecast." In *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, 459–464. Vancouver, Canada: Association for Computational Linguistics.

Sap, Maarten, Prasettio, Marcella Cindy, Holtzman, Ari, Rashkin, Hannah, and Choi, Yejin. 2017. "Connotation Frames of Power and Agency in Modern Films." In *Proceedings of the 2017 Conference on Empirical Methods in Natural Language Processing*, 2329–2334. Copenhagen, Denmark: Association for Computational Linguistics.

Seddon, Toby. 2008. "Drugs, the Informal Economy and Globalization." *International Journal of Social Economics* 35 (10): 717–28.

Wikipedia-API: Python Wrapper for Wikipedia. Python. PyPI, <https://github.com/martin-majlis/Wikipedia-API>. Accessed 24 Apr. 2026, OS Independent.

Zieger, Susan. 2008. *Inventing the Addict: Drugs, Race, and Sexuality in Nineteenth-Century British and American Literature*. Amherst: University of Massachusetts Press.